

Resimulation Scope Module (SRSM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

Parent Standard: Simulation Lifecycle & Resimulation Standard (SLRS)

Module: SRSM — Resimulation Scope Module

Operational Layer: Resimulation Scope Governance Layer

Category: Governance & Enforcement

Subcategory: Simulation Governance

Type: Core Module

Governed Space: Resimulation Scope

Origin ID: SIMULOS-SLRS-SRSM-012.4

Version: 1.0

Status: Canonical · Open Standard

Effective Date: 20 August 2026

Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS® · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · RIS™

AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodology Intellectual Property Protection

Canonical Language: English

Canonical Definition

Resimulation Scope Module (SRSM) is the fourth Core Module of the Simulation Lifecycle & Resimulation Standard (SLRS) within SIMULOS® — Simulation Governance Architecture.

SRSM governs the determination of what must be resimulated after a governed Resimulation Trigger has been established.

The Module determines the minimum sufficient and methodologically justified resimulation boundary across affected Simulation Objects, configurations, representations, assumptions, models, environments, inputs, scenarios, executions, outputs, correlations, and simulation-derived evidence.

SRSM prevents two opposite governance failures:

insufficient resimulation, where the renewed simulation scope fails to cover materially affected dependencies;

and

unnecessary resimulation, where unaffected simulation spaces are repeated without a methodological reason.

Its task is therefore:

to translate a Resimulation Trigger into an explicit, dependency-aware, traceable, and proportionate Resimulation Scope.

Architectural Position

SRSM is the fourth module within SLRS.

It follows:

SRGM — Resimulation Trigger Module

and precedes:

SRTM — Simulation Retirement & Transition Module.

The sequence is:

Lifecycle Monitoring → Drift Recognition → Resimulation Trigger → Resimulation Scope → Retirement & Transition.

SRGM establishes:

whether resimulation is required.

SRSM establishes:

what must be resimulated.

SRTM subsequently governs whether a simulation should continue through its lifecycle, transition toward a replacement, or be retired.

Internal Flow Position

Governed Resimulation Trigger Record → Affected Simulation Basis Identification → Dependency Propagation Analysis → Affected Object / Configuration Identification → Affected Scenario Space Identification → Affected Model / Environment Identification → Affected Input Basis Identification → Affected Execution Requirements → Affected Output / Correlation / Evidence Identification → Scope Sufficiency Assessment → Resimulation Scope Determination → Governed Resimulation Scope Record → Resimulation Lifecycle Re-Entry / SRTM Operational Space

SRSM governs the space between:

a confirmed need to resimulate

and

the definition of the simulation lifecycle elements that must be reopened, updated, repeated, reconstructed, or reassessed.

This operational space includes: resimulation scope identification; affected simulation component identification; dependency propagation; impact boundaries; scenario re-entry; configuration re-entry; model re-entry; input re-entry; assumption re-entry; representation re-entry; execution re-entry; output reassessment; correlation reassessment; evidence consequence

identification; partial resimulation; targeted resimulation; campaign resimulation; comprehensive resimulation; scope expansion; scope reduction; unaffected-space preservation; scope sufficiency; scope documentation.

Operational Scope

SRSM applies whenever a Resimulation Trigger requires some portion of an existing simulation basis to be reconsidered or executed again.

It may govern: one simulation run; one parameter range; one scenario; one scenario family; one Simulation Object configuration; one subsystem; one model component; one environmental condition; one input distribution; one failure mode; one mission phase; one behavioral domain; one simulation campaign; multiple interconnected campaigns; the complete simulation lifecycle.

It applies across engineering, scientific, physical, digital, hybrid, mission, safety, infrastructure, economic, AI, autonomous-system, digital-twin, and future simulation environments.

SRSM remains technology-neutral. Trigger ≠ Scope

SRSM establishes the fundamental distinction:

Resimulation Trigger ≠ Resimulation Scope

A trigger establishes that the existing simulation basis requires renewed simulation activity.

It does not establish how much must be repeated.

A material change in one subsystem may require:

targeted resimulation

rather than:

complete campaign repetition.

Conversely, a seemingly localized change may propagate through dependencies and require much broader resimulation than its initial location suggests.

Therefore:

The size of the originating change does not determine the size of the required resimulation scope.

Minimum Sufficient Scope Principle

SRSM seeks the Minimum Sufficient Resimulation Scope.

This means:

the smallest resimulation boundary that fully captures all materially affected simulation dependencies necessary to restore the governed simulation basis for the intended use.

“Minimum” prevents unnecessary repetition.

“Sufficient” prevents under-scoping.

The objective is not computational minimization at any cost.

The objective is methodologically justified proportionality.

Affected Simulation Basis

Every Resimulation Trigger must be mapped to the simulation basis it potentially affects.

The affected basis may include: Simulation Object; object version; object configuration; simulation purpose; scope; exclusions; reality representation; abstraction; fidelity; assumptions; model structure; environment; parameters; inputs; data; temporal conditions; scenarios; scenario coverage; execution configuration; outputs; correlation relationships; evidence objects.

The scope cannot be determined reliably until these dependencies are visible.

Dependency Propagation Principle

A change may propagate beyond the simulation element where it originated.

For example:

Input Change → affects Model Behavior → affects Scenario Outcomes → affects Outputs → affects Simulation-to-Reality Correlation → affects Simulation Evidence.

Or:

Object Configuration Change → affects Reality Representation → affects Assumptions → affects Model Parameterization → affects Scenario Relevance → affects Execution → affects Outputs.

SRSB therefore requires dependency-aware scope determination. Local Change ≠ Local Scope

A localized technical change does not guarantee a localized resimulation requirement.

A single changed parameter may influence: multiple model components; multiple scenario families; downstream control behavior; interaction effects; failure thresholds; evidence already derived from previous executions.

Conversely, a large physical modification may have no material effect on a narrowly defined simulation question.

Scope must follow material simulation dependency, not superficial change size.

Resimulation Scope Levels

SRSB may classify scope into governed levels. 1. Execution-Level Resimulation

Repeat one or more executions without materially changing the upstream simulation basis.

Appropriate where the issue concerns: execution anomaly; stochastic repetition; failed run; incomplete execution; runtime instability.

2. Scenario-Level Resimulation

Repeat or expand specific scenarios or scenario families.

Appropriate where: new edge cases emerge; scenario conditions change; coverage gaps are discovered; one scenario class becomes materially relevant.

3. Input-Level Resimulation

Renew simulation where input or data conditions materially change.

This may require: new datasets; updated distributions; changed temporal inputs; changed uncertainty ranges.

4. Configuration-Level Resimulation

Repeat simulations for affected Simulation Object, model, or environment configurations. 5. Model-Level Resimulation

Reopen the simulation model or environment basis before execution.

Appropriate where: model structure changes; model limitations are discovered; environment behavior changes; parameterization must be reconstructed.

6. Representation-Level Resimulation

Reopen the representation of reality itself.

Appropriate where previously simplified, excluded, or inaccurately represented reality becomes materially relevant. 7. Comprehensive Resimulation

Re-enter multiple or all materially relevant simulation lifecycle stages.

This may be necessary where change is systemic, dependencies are extensive, or the previous simulation basis can no longer support meaningful partial reuse.

These levels may overlap.

They are governance classifications, not rigid technical implementation categories.

Partial Resimulation

SRSM permits partial resimulation where the unaffected simulation basis remains legitimate.

A partial scope should establish: what remains unchanged; why it remains reusable; what must be updated; what must be repeated; what downstream results become affected; what previous evidence remains applicable; what evidence requires reassessment.

Partial resimulation must not become an assumption of convenience.

Its sufficiency must be demonstrable.

Targeted Resimulation

Targeted resimulation may focus computational and analytical resources upon the specific conditions affected by the Trigger.

Examples include: one failure mode; one sensor condition; one mission phase; one environmental range; one scenario family; one subsystem interaction; one parameter region.

Targeted resimulation can dramatically reduce unnecessary simulation effort while preserving methodological sufficiency.

Comprehensive Resimulation

Comprehensive resimulation may be required where: Simulation Object identity materially changes; reality representation becomes structurally insufficient; foundational assumptions fail; model architecture changes substantially; multiple scenario families become affected; dependencies cannot be reliably isolated; correlation deteriorates broadly; accumulated drift becomes systemic; previous evidence can no longer be cleanly separated from the affected basis.

Comprehensive resimulation should result from governed scope analysis rather than default organizational habit.

Scope Expansion Principle

An initially narrow Resimulation Scope may need to expand.

During renewed simulation activity, organizations may discover: additional affected dependencies; new failure modes; broader scenario impact; unexpected sensitivity; new model interactions; wider correlation deterioration; additional evidence consequences.

SRSM therefore allows:

Initial Scope → New Information → Scope Reassessment → Expanded Scope.

A scope determination must not become immutable when new material information emerges.

Scope Reduction Principle

A provisional scope may also be reduced where evidence establishes that suspected dependencies are not materially affected.

Reduction must be justified.

It should identify: the excluded simulation space; the evidence supporting exclusion; the dependency analysis; remaining uncertainty.

Scope Reduction ≠ Convenience-Based Exclusion.

Unchanged Basis Preservation

SRSM should preserve simulation components that remain legitimately reusable.

This may include: unaffected scenarios; unchanged input sets; unaffected model components; stable environmental configurations; previous execution records; unaffected outputs; still-applicable correlation evidence.

This prevents simulation governance from creating unnecessary computational waste.

However, reuse requires a governed basis.

Previously Simulated $\neq$ Automatically Reusable.

Scenario Scope

Where the Trigger affects scenario coverage, SRSM determines which scenario space must be reopened.

This may include: expected conditions; variations; edge cases; stress conditions; uncertainty conditions; interaction scenarios; newly observed scenarios; previously excluded scenarios.

SSS continues to govern scenario design.

SRSM determines which scenario governance must be re-entered because of the Trigger.

Model & Environment Scope

Where the Trigger affects the simulation model or environment, SRSM determines which model or environmental elements require renewed governance.

Possible consequences include: model component revision; parameterization revision; environment reconfiguration; boundary interaction reassessment; version change; dependency reconstruction.

SMES governs these elements when re-entered.

SRSM governs the requirement to reopen them.

Input & Data Scope

Where simulation inputs are affected, SRSM determines which input basis must be reconsidered.

Possible scope may include: one variable; one dataset; one distribution; one time period; one environmental source; one uncertainty range; the complete input basis.

SIDS governs the renewed input and data conditions.

SRSM defines the affected boundary.

Assumption Scope

A changed assumption may propagate through many simulation layers.

SRSM must determine: which assumption changed; which model components depend upon it; which scenarios rely upon it; which executions used it; which outputs were affected; which correlation conclusions depend upon it; which evidence objects inherit the dependency.

SAS governs the assumption itself.

SRSM governs the lifecycle scope affected by its change.

Output and Evidence Consequence Scope

Resimulation scope does not stop at new execution.

Where prior outputs or simulation-derived evidence depend upon the affected simulation basis, SRSM should identify them for downstream reconsideration.

This may require renewed governance through:

SOGS → SRCS → SERBS.

A new simulation execution may therefore invalidate, supersede, narrow, strengthen, or leave unchanged different portions of the previous evidence basis.

Scope Sufficiency Assessment

Before a Resimulation Scope is accepted, SRSM should determine whether the proposed scope is sufficient to address the Trigger.

The assessment may ask: Have all material dependencies been identified? Are affected scenarios included? Are changed assumptions covered? Are model consequences captured? Are affected inputs reconsidered? Are downstream outputs identified? Is correlation reassessment required? Is existing evidence affected? Are excluded simulation spaces demonstrably unaffected? Could residual uncertainty materially undermine the intended use?

The purpose is to prevent a technically completed resimulation from remaining methodologically incomplete.

Scope Uncertainty

Sometimes the required scope cannot initially be determined with confidence.

SRSM may therefore establish:

Provisional Resimulation Scope

where uncertainty remains.

The provisional scope may be expanded or reduced as evidence emerges.

Uncertainty about scope should be documented rather than hidden through arbitrary boundary selection.

Resimulation Re-Entry Principle

SRSM does not require every Resimulation Trigger to restart SIMULOS® from Standard 1.

Instead, it determines the earliest materially affected governance point.

Examples:

Execution anomaly → re-enter at SES.

New scenario → re-enter at SSS.

Input distribution shift → re-enter at SIDS.

Model revision → re-enter at SMES.

Assumption failure → re-enter at SAS.

Reality representation deficiency → re-enter at RRS.

Simulation Object material change → potentially re-enter at SOS.

This enables SIMULOS® to operate recursively without forcing unnecessary full-lifecycle repetition.

Lifecycle Propagation

Once the re-entry point is established, downstream governance should be reconsidered wherever materially affected.

For example:

SAS Re-Entry → SMES → SIDS where affected → SSS → SES → SOGS → SRCS → SERBS.

Not every downstream element must necessarily change.

But every materially dependent downstream element must be considered.

SRSM Boundary with SRGM

SRGM determines:

whether resimulation is required.

SRSM determines:

what must be resimulated.

Therefore:

Trigger ≠ Scope.

SRSM must not redefine the original Trigger merely to reduce or expand workload.

SRSM Boundary with SSS

SRSM may determine that a scenario space must be reopened.

SSS governs the actual identification, variation, edge-case design, stress conditions, and coverage of those scenarios.

SRSM Boundary with SMES

SRSM may determine that the model or environment must be reopened.

SMES governs model structure, environmental configuration, boundary interactions, parameterization, and environment versioning.

SRSM Boundary with SIDS

SRSM may identify affected input and data spaces.

SIDS governs those inputs once simulation governance re-enters that layer.

SRSM Boundary with SES

SRSM determines which executions must occur again.

SES governs how those simulations are instantiated, controlled, observed, repeated, and documented.

SRSM Boundary with SOGS

SRSM may determine which prior or newly produced outputs require renewed governance.

SOGS governs output identification, classification, inference, limitations, and documentation.

SRSM Boundary with SRCS

SRSM may determine that simulation-to-reality correlation must be reassessed because the simulation basis has materially changed.

SRCS governs the actual correlation process.

SRSM Boundary with SERBS

SRSM identifies simulation-derived evidence potentially affected by resimulation.

SERBS determines the qualification, weight, reliance boundary, limitations, and validation interface of that evidence.

SRSM Boundary with VALIDOS®

SRSM determines the simulation-specific scope of resimulation.

It does not determine the broader scope of revalidation.

A VALIDOS® revalidation process may require additional evidence or activities outside simulation.

SRSM Boundary with ORA™

Operational Reality may provide changes that influence scope determination.

ORA™ continues to govern Operational Reality itself.

SRSM governs only the resulting simulation lifecycle re-entry boundary.

SRSM Boundary with INTEGROS®

SRSM does not govern general integrity.

Integrity issues may create dependencies affecting resimulation scope, but integrity governance remains under INTEGROS®.

SRSM Boundary with ADIT®

ADIT® may audit whether the basis, exclusions, dependency analysis, scope determination, and resulting resimulation records are preserved and verifiable.

SRSM governs the scope determination itself.

Minimum Implementation Framework

A minimum SRSM implementation should establish: 1. Trigger Intake

Receive the governed Resimulation Trigger and its supporting basis. 2. Affected Basis Identification

Identify the simulation lifecycle elements directly affected by the Trigger. 3. Dependency Mapping

Determine material upstream, downstream, and cross-component dependencies. 4. Re-Entry Point Determination

Identify the earliest materially affected SIMULOS® governance layer. 5. Scope Classification

Determine whether execution-level, scenario-level, input-level, configuration-level, model-level, representation-level, or comprehensive resimulation is required. 6. Unaffected Basis Assessment

Identify previous simulation elements that remain legitimately reusable. 7. Downstream Consequence Assessment

Identify outputs, correlation relationships, evidence objects, and reliance boundaries requiring reconsideration. 8. Scope Sufficiency Assessment

Determine whether the proposed boundary captures all material consequences of the Trigger. 9. Scope Adjustment Control

Permit governed expansion or reduction where new information materially changes the required boundary. 10. Resimulation Scope Record

Document the final or provisional Resimulation Scope, exclusions, dependencies, re-entry point, and downstream consequences. Use Case 1 — Space Mission Simulation: Flight Software Change Before Launch

A spacecraft has completed a very large pre-launch simulation campaign covering: ascent; navigation; orbital operations; descent; landing; communication failures; propulsion anomalies; contingency scenarios.

A late flight-software update modifies the guidance algorithm used during descent.

SRGM has already established a Priority Resimulation Trigger.

The organization now faces a major practical question:

Must the complete mission simulation campaign be repeated?

SRSB maps the change through the simulation dependency structure.

The updated guidance algorithm affects:

Descent Control → Navigation Interaction → Propulsion Commands → Landing Trajectory → Fuel Consumption → Landing-Site Dispersion → Failure and Contingency Scenarios → Mission Outputs → Relevant Simulation Evidence.

But the change may have no material relationship to unrelated orbital communication scenarios or other unaffected mission phases.

SRSB therefore establishes a targeted but dependency-complete Resimulation Scope.

The organization re-enters the simulation lifecycle at the affected model/configuration layer, reconstructs the relevant scenario space, repeats the affected execution campaign, regenerates governed outputs, and reassesses relevant simulation-to-reality correlation and evidence.

This avoids both extremes:

repeating the entire mission campaign unnecessarily

and

testing only the modified algorithm while missing its downstream mission interactions.

The governing principle is:

Resimulate the dependency chain affected by the change, not merely the component where the change originated. Use Case 2 — Autonomous Robot: Rare Failure Reveals a Missing Scenario Domain

A mobile autonomous robot has successfully completed millions of simulations before deployment.

Operational observation later reveals failures occurring when three conditions

combine:

low-angle sunlight + partially contaminated sensor surface + unusual pedestrian movement.

SRGM determines that the new failure pattern creates a Resimulation Trigger.

SRSM must now determine the required scope.

Repeating the complete historical campaign may consume enormous computational resources while adding little information.

Repeating only the exact failed event may be too narrow because the failure could belong to a broader interaction domain.

SRSM therefore traces the affected dependencies:

New Reality Condition → Reality Representation → Sensor Model → Environmental Inputs → Scenario Family → Autonomous Behavior → Execution → Outputs → Sim-to-Real Correlation → Existing Evidence.

The resulting scope may require: updating the representation of contaminated sensors; expanding relevant lighting ranges; creating variations of human movement; testing combinations rather than isolated conditions; rerunning the affected behavioral scenario family; reassessing outputs; recalculating relevant correlation; reconsidering simulation evidence previously used for that operational domain.

Unaffected simulation domains remain reusable where their independence is established.

The organization therefore avoids the misleading choice between:

rerun everything

and

rerun only the failure.

SRSM establishes the third option:

rerun everything materially connected to the failure, and preserve everything demonstrably unaffected.

Governance Outputs

SRSM may produce: Resimulation Scope Assessment Affected Simulation Basis Map
Simulation Dependency Map Resimulation Re-Entry Point Resimulation Scope
Classification Affected Scenario Scope Affected Input Scope Affected Model &
Environment Scope Affected Configuration Scope Required Execution Scope
Downstream Output Reassessment Scope Correlation Reassessment Requirement
Affected Simulation Evidence Register Unaffected Simulation Basis Statement
Scope Exclusion Record Scope Sufficiency Assessment Provisional Resimulation
Scope Final Resimulation Scope Governed Resimulation Scope Record

Enterprise Application

SRSM gives organizations a method for answering one of the most expensive questions in simulation lifecycle management:

When something changes, how much simulation must actually be repeated?

Without governed scope determination, organizations may either repeat massive simulation campaigns unnecessarily or perform narrow retesting that fails to capture downstream dependencies.

SRSM enables simulation-intensive organizations to determine: the earliest lifecycle point requiring re-entry; which scenarios must be repeated; which model components require revision; which inputs must be updated; which configurations are affected; which executions must be repeated; which previous outputs remain usable; which correlation relationships require reassessment; which simulation-derived evidence has become affected; which unaffected simulation assets may legitimately be preserved.

This becomes especially important where simulation campaigns contain millions or billions of executions.

The objective is not simply to reduce simulation cost.

It is to ensure that computational effort follows methodological necessity .

Foundational Principles

Resimulation Trigger does not determine Resimulation Scope.

The size of a change does not determine the size of its simulation consequence.

Local change does not necessarily imply local resimulation.

Resimulation scope must follow material dependency.

The minimum scope must still be sufficient.

Unaffected simulation assets should not be discarded merely because another part of the simulation changed.

Previously simulated does not mean automatically reusable.

A resimulation may re-enter SIMULOS® at any materially affected lifecycle layer.

Downstream outputs and evidence must be reconsidered where their simulation basis has changed.

Scope may expand or contract when new information changes the known dependency boundary.

Resimulation should repeat neither less nor more than the materially affected simulation basis requires.

Canonical Closing Statement

Resimulation Scope Module establishes the scope-determination layer of the Simulation Lifecycle & Resimulation Standard by transforming a governed Resimulation Trigger into an explicit, dependency-aware, proportionate, sufficient, and traceable Resimulation Scope. Through governance of affected simulation bases, dependency propagation, lifecycle re-entry points, scenario scope, input scope, model and environment scope, configuration scope, execution requirements, downstream output consequences, correlation reassessment, evidence consequences, unaffected-basis preservation, scope sufficiency, and scope adjustment, SRSM prevents both insufficient resimulation and unnecessary repetition of unaffected simulation spaces. By separating the decision that resimulation is required from the decision concerning what must be resimulated, SRSM enables SIMULOS® to operate recursively across individual executions, targeted scenario families, model revisions, configuration changes, large simulation campaigns, and complete lifecycle reconstruction. Resimulation should not restart more of the simulation lifecycle than necessary, but it must reopen every materially affected dependency required to restore a legitimate governed simulation basis.

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>